

Circulum Vita

Dr. Viswanath Chinthapenta

Associate Professor

MAE-512

Indian Institute of Technology Hyderabad,

Department of Mechanical and Aerospace Engineering,

Kandi, Sangareddy, Telangana, India-502285.

(E): viswanath@mae.iith.ac.in

(AE): viswanath.chinthapenta@gmail.com

(O): 040-2301-6661

(M): +91-8790128695

[Google Scholar: qVCJiNIAAAAJ](#)

[Scopus Author ID: 36994500100](#)

[ORCID: 0000-0001-9038-6274](#)



Educational Qualification:

Degree	Year	Subject	Institute/University
Ph.D.	2012	Solid Mechanics	Brown University, US
M.E.	2005	Aerospace Engineering	IISc, Bangalore
B.E.	2003	Civil Engineering	OU/ MVSRR Engg. College

Professional Experience:

S No	Position	Organization	Period
6	Assoc. Prof.	Department of MAE, IIT Hyderabad	2021-Till date
5	Asst. Prof	Department of MAE, IIT Hyderabad	2015-2021
4	Visiting Asst. Prof.	Department of MAE, IIT Hyderabad	2013-2015
3	Post-Doctoral	Dept. of Aerospace Engineering, IISc Bangalore	2012-2013
2	Research Intern	RHS team, Life Sciences Group, DSS, USA, Providence	2012-2012
1	Engineer	JFWTC, GE, Whitefield, Bangalore, India	2005-2006

Specialization and Expertise:

Computational Solid Mechanics, Additive Manufacturing, and Bio-Mechanics

Awards and Distinctions:

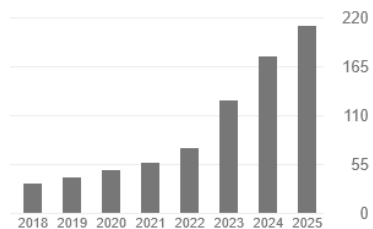
SNo	Award/Distinctions
12	IIM-SJEF travel grant to attend AAMS 2025 to PhD Student, Mr Sahil Dhiman
11	3 rd Best Paper Award, ICLPRP 2025 to PhD Student, Mr Aryendra Singh
10	Bursar Award and ANRF travel grant APICAM 2025 to PhD Student, Mr Sahil Dhiman
9	Best Paper Award PAMFP-2024 to PhD Student, Mr. Sahil Dhiman

8	Best Paper Award, EECSS-2024 to Mr Sriram HK
7	IITH Research Excellence Award 2024 to PhD Student, Mr. Himanshu Gururani
6	Prime Minister's Office (PMO) fellowship 2018 to PhD student, Mr. Himanshu G
5	Best Poster Award SICE-2016 to ME student, Ms. Mohini Kale
4	Graduate Research Assistantship, Brown University 2007-2012
3	Graduate Fellowship, Brown University 2006-2007
2	MHRD Graduate Assistance, IISc Bangalore, 2003-2005
1	All India Rank 3, GATE-2003, Civil Engineering

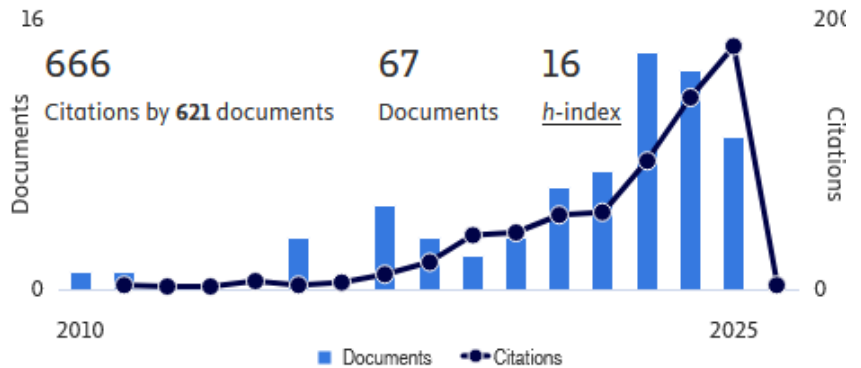
Research and Development:

a) Overall Profile:

	All	Since 2020
Citations	820	695
h-index	17	17
i10-index	23	22



[Google Scholar: qVCJiNIAAAAJ](https://scholar.google.com/citations?user=qVCJiNIAAAAJ)



[Scopus Author ID: 36994500100](https://scopus.com/authid/detail/authid/36994500100)

[ORCID: 0000-0001-9038-6274](https://orcid.org/0000-0001-9038-6274)

b) Publications (with IF/CS):

SNo	Journal Publications	IF	CS
50	G. Satya Sasank, Aryendra Singh, P. Prem Kiran, Viswanath Chinthapenta , T. Jayakumar, Phani Mylavarapu*. An Enhanced Fabbro Model for Water-Confined Plasma Pressure and Its Application in Laser Shock Peening: Influence of Laser Intensity and Coupling Efficiency. Optics & Laser Technology, Volume 192, Part F, December 2025, 114062. doi.org/10.1016/j.optlastec.2025.114062	5	9.2
49	P Mahesh, V Chinthapenta , G Raju, M Ramji*. Progressive damage analysis of open-hole CFRP laminates under combined tension-shear loading. Composite Structures, Volume 373, 2025, 119676. doi.org/10.1016/j.compstruct.2025.119676	7.1	14.8
48	S Chalasani, A Singh, PS Reddy, V Chinthapenta *. Enhancement of Thrust Vector Control by Composite Reinforcements in Conical Elastomeric Bearing. Composites Part B: Engineering, Volume 307, 2025, 112970. doi.org/10.1016/j.compositesb.2025.112970	14.2	25.6
47	Nikesh Chelimilla, Viswanath Chinthapenta , Srikanth Korla*. Audio feature augmentation for bolt looseness classification in data-deficient scenarios using machine learning. Nondestructive Testing and Evaluation. Volume 40, 2025. doi.org/10.1080/10589759.2024.2405884	4.2	4.4
46	S Dhiman, M Brandt, D Fabijanic, V Chinthapenta *, W Xu*. Microstructural control across multiple length scales in additively manufactured Ti-6Al-4V via cyclic heat treatments, Acta Materialia, Volume 297, 2025, 121372. doi.org/10.1016/j.actamat.2025.121372	9.3	15.4
45	A Chaudhary, R Mishra, C Ramesh, V Chinthapenta , M Imam*, Y Sun. Experimental and numerical investigation of laser-FSW hybrid welding technique for high strength materials and thick aluminium welds. Welding International, 1-31. doi.org/10.1080/09507116.2025.2527750	1.2	1.3
44	A Sabale, MS Rizvi, V Chinthapenta , A Eranki*. Effect of Focused Ultrasound on Shearwave Production in a Hyperelastic Media. Biomechanics and Modeling in Mechanobiology, Volume 24, pages 1279–1294, 2025. doi.org/10.1007/s10237-025-01967-2	3.2	6.4

43	Chandan Kumar, Sarathchandra Reddy, Nandha Kumar Eswaramoorthy, Praveen Ravanappa, Viswanath Chinthapenta , Dheepa Srinivasan*, Praveen Kumar*. Microstructure optimization for improving creep resistance of additively manufactured Ni-based superalloy IN939 through heat treatment. Journal of Materials Science. 60, pp. 1545-1560, 2025. doi.org/10.1007/s10853-024-10337-9	4.3	7.6
42	A Khalad, G Telasang, K Kadali, PN Zhang, W Xu, V Chinthapenta *. A generalized machine learning framework for data-driven prediction of relative density in laser powder bed fusion parts. The International Journal of Advanced Manufacturing Technology 135 (9), 4147 – 4167, 2024. doi.org/10.1007/s00170-024-14735-w	3.5	5.9
41	SA Patil, V Chinthapenta , SN Khaderi, M Ramji*. Experimental and numerical estimation of complex stress intensity factor for the completely debonded anti-crack embedded into a weak matrix using domain integral method. Theoretical and Applied Fracture Mechanics, Volume 133, Part B, 2024, 104642. doi.org/10.1016/j.tafmec.2024.104642	5.6	8.4
40	Sahil Dhiman, Viswanath Chinthapenta *, Milan Brandt, Daniel Fabijanic, Wei Xu*. Microstructure control in additively manufactured Ti-6Al-4V during high-power laser powder bed fusion. Additive Manufacturing. 96, 104573, 2024. doi.org/10.1016/j.addma.2024.104573	11.1	20
39	P Mahesh, Viswanath Chinthapenta , Gangadharan Raju, M Ramji*. Classification of damage modes in scaled open-hole composite laminates under combined tension-shear loading using acoustic emission technique. Engineering Fracture Mechanics, 308, pp 10390, 2024. doi.org/10.1016/j.engfracmech.2024.110390	5.3	8.6
38	Himanshu Gururani, Sai Naga Sri Harsha Chittajallu, Minal Doulatramani, Viswanath Chinthapenta , Sayan Basu, M Ramji*. Collagen imaging reveals synergistic effects of sutures and host-donor misalignment on topographical irregularities in penetrating keratoplasty. PloS One, 19(8), e0308204, 2024. doi.org/10.1371/journal.pone.0308204	3.2	5.4
37	Viswanath Chinthapenta *, Shailendra Joshi, Srinivasan Chandrasekar. Introduction to the special issue on structural integrity. International Journal of Fracture. 247(2), pp 133-134, 2024. doi.org/10.1007/s10704-024-00804-4	2.5	4.5
36	Nikesh Chelimilla, Viswanath Chinthapenta , Srikanth Korla*. Addressing data scarcity using audio signal augmentation and deep learning for bolt looseness prediction. Smart Materials and Structures. 33(8), 085012, 2024. doi.org/10.1088/1361-665X/ad5c24	3.8	6.1
35	S. N. S. H. Chittajallu, Himanshu Gururani, Saumya Jakati, Sayan Basu, Pravin Krishna Vaddavalli*, Kwong Ming Tse, Viswanath Chinthapenta *. Investigation of Mechanical Strength and Structure of Corneal Graft-Host Junction. Heliyon. 10(10) e30871, 2024. doi.org/10.1016/j.heliyon.2024.e30871	3.6	4.1
34	Mahesh P, Viswanath Ch , Gangadharan Raju, and Ramji M*. Experimental investigation on open-hole CFRP laminate under combined loading using acoustic emission and digital image correlation. Theoretical and Applied Fracture Mechanics. 130, April 2024 Article number 104300. doi.org/10.1016/j.tafmec.2024.104300	5.6	8.4
33	Singh A, Chinthapenta V , Mylavarapu P*. Efficacy of Laser Shock Peening Postmilling: A Semi-numerical Study. Journal of Materials Engineering and Performance, 33(8), pp. 4106-4113, 2024. doi.org/10.1007/s11665-024-09241-9	2.3	4.4
32	H Gururani, SNSH Chittajallu, M Doulatramani, R Manoharan, S Basu, Viswanath Ch *. Intraoperative collagen imaging of sutured cornea: A way towards managing postpenetrating keratoplasty astigmatism. Medical Engineering & Physics 123, 104076, 2024. doi.org/10.1016/j.medengphy.2023.104076	2.3	4.2
31	Nikesh C, Viswanath Ch , Naresh K, Srikanth K*. Review on recent advances in structural health monitoring paradigm for looseness detection in bolted assemblies. Structural Health Monitoring, Volume 22, Issue 6, Pages 4264 – 4304, November 2023. doi.org/10.1177/14759217231158540	7.3	14.5
30	Patil S, Khaderi S N, Ramji M, Viswanath Ch *. Full Field Solution for Remotely Loaded One Side Completely Debonded Short Rigid Line Inclusion Embedded in Soft Matrix: Two-Dimensional Analytical and Experimental Insights. Journal of Applied Mechanics, Transactions ASME. Volume 90, Issue 101, October 2023 Article number 101006. doi.org/10.1115/1.4062771	2.8	5
29	H Gururani, SNSH Chittajallu, R Manoharan, S Basu, V Chinthapenta *. Identification of subject-specific fibrillar disposition in healthy rabbit cornea through birefringence analysis. Optics and Lasers in Engineering 169, 107747, 2023. doi.org/10.1016/j.optlaseng.2023.107747	3.7	7.1

28	SNSH Chittajallu, H Gururani, KM Tse, SN Rath, S Basu, V Chinthapenta* . Investigation of microstructural failure in the human cornea through fracture tests. <i>Scientific Reports</i> 13 (1), 13876, 2023. doi.org/10.1038/s41598-023-40286-3	4.3	6.7
27	G Gulivindala, MK Karanam, KM Tse, V Chinthapenta* . Influence of material anisotropy on void coalescence by necking for face-centred cubic single crystals. <i>Materials Today Communications</i> 35, 106010, 2023. doi.org/10.1016/j.mtcomm.2023.106010	4.5	5.8
26	MAA Anshari, R Mishra, M Imam*, D Pandit, KK Jha, R Kesharwani, M Kar, V Chinthapenta . Comparison of the Microstructures and Mechanical Properties in the Overlapping Region of Low Carbon Steel Additive Bead Fabricated by WAAM and FSP. <i>Metallurgical and Materials Transactions A</i> 54 (3), 869-895, 2023. doi.org/10.1007/s11661-022-06934-5	2.8	4.4
25	R Mishra*, A Pillai, M Imam, V Chinthapenta , KP Vineesh. Experimental and Numerical Investigation of Double Pass Overlapping Beads by Twin-Wire Welding Additive Manufacturing Process. <i>Transactions of the Indian Institute of Metals</i> 76 (2), 297-313, 2023. doi.org/10.1007/s12666-022-02666-7	1.6	2.8
24	H Gururani, S Chittajallu, M Ramji, S Basu, V Chinthapenta* . An In-Vitro Investigation on the Birefringence of the Human Cornea Using Digital Photoelasticity. <i>Experimental Mechanics</i> , 2023, 63(2), pp. 205–219. doi.org/10.1007/s11340-022-00910-1	2.7	4.5
23	S Potukuchi, V Chinthapenta , G Raju*. A review of NDE techniques for hydrogels. <i>Nondestructive Testing and Evaluation</i> 38 (1), 1-33, 2023. doi.org/10.1080/10589759.2022.2144304	4.2	4.4
22	GV Krishna, V Narayanamurthy*, C Viswanath . Buckling behaviour of FRP strengthened cylindrical metallic shells with cut-outs. <i>Composite Structures</i> , 116176 2022. doi.org/10.1016/j.compstruct.2022.116176	7.1	14.8
21	G Gulivindala, MK Karanam, R Ramadurai, V Chinthapenta* . Indentation-Based Fracture Toughness Estimation of Barium Titanate Thin Film Using Experiments and Simulations. <i>Thin Solid Films</i> , 2022, 755, 139325. doi.org/10.1016/j.tsf.2022.139325	2	3.9
20	Sai Naga Sri Harsha Chittajallu, Kwong Ming Tse*, Ashutosh Richhariya, Viswanath Chinthapenta* . A Review on Damage and Rupture Modeling for Soft Tissues. <i>Bioengineering</i> , 2022, 9(1), 26, 2022. doi.org/10.3390/bioengineering9010026	3.7	5.3
19	GV Krishna, V Narayanamurthy*, C Viswanath . Effectiveness of FRP strengthening on buckling characteristics of metallic cylindrical shells. <i>Composite Structures</i> , 2021, 262, 113653, 2021. doi.org/10.1016/j.compstruct.2021.113653	7.1	14.8
18	H Gururani, A Richhariya, M Ramji, V Chinthapenta* . Implications of the structure-property relationship on the optomechanical characterization of the cornea: A review. <i>Optik. Volume 232 April 2021 Article number 166529</i> . doi.org/10.1016/j.ijleo.2021.166529	3.1	8.3
17	MK Karanam, G Gulivindala, VR Chinthapenta . Effect of anisotropy on the ductile fracture in metal reinforcements of brittle matrix composites. <i>Theoretical and Applied Fracture Mechanics</i> Volume 112, 2021, 102923. doi.org/10.1016/j.tafmec.2021.102923	5.6	8.4
16	MK Karanam, V Chinthapenta* . Void growth and morphology evolution during ductile failure in an FCC single crystal. <i>Continuum Mechanics and Thermodynamics</i> , Volume 33, Pages 497 – 513, 2021. doi.org/10.1007/s00161-020-00922-z	2.1	5.2
15	MAA Anshari, M Imam*, MZK Yusufzai, V Chinthapenta , R Mishra. Stir zone anisotropic work hardening behavior in friction stir-processed EN8 medium carbon steel. <i>Materials Science and Engineering: A</i> , 2021, 805, 140582. doi.org/10.1016/j.msea.2020.140582	7	12.7
14	T Chaitanya Sagar, Viswanath Chinthapenta* , and MF Horstemeyer. Effect of defect-guided out-of-plane deformations on the mechanical properties of graphene. <i>Fullerenes, Nanotubes and Carbon Nanostructures</i> , Volume 29, 83-99, 2021. doi.org/10.1080/1536383X.2020.1813720	2.2	4
13	G. Vamsi Krishna, V. Narayanamurthy*, and C. Viswanath . Modeling of buckling characteristics of the metal-FRP hybrid cylinder. <i>International Journal of Composite Structures</i> , 250, 112505, 2020. doi.org/10.1016/j.compstruct.2020.112505	7.1	14.8
12	T Chaitanya Sagar, Viswanath Chinthapenta* . Effect of the substitutional and vacancy defects on 2D h-Boron Nitride. <i>Journal of Molecular Modelling</i> , 26 (8), 192, 2020. doi.org/10.1007/s00894-020-04452-y	2.2	3.8

11	S Matta, NR Kolanu, V Chinthapenta , CM Manjunatha, M Ramji*. Progressive damage analysis of adhesively bonded patch-repaired carbon fiber-reinforced polymer specimen under compression involving cohesive zone model. International Journal of Damage Mechanics, 28(10), 1457-1489, 2019. doi.org/10.1177/1056789519832062	3.6	9.1
10	Imam Murshid*, Sun Yufeng, Fujii Hidetoshi, Ninshu MA, Tsutsumi Seiichiro, Ahmed Shuja, Chintapenta Viswanath , Murakawa Hidekazu. Deformation characteristics and microstructural evolution in friction stir welding of thick 5083 aluminum alloy. The International Journal of Advanced Manufacturing Technology, 99, 663-681, 2018. doi.org/10.1007/s00170-018-2521-9	3.5	5.9
9	Matta Seshadri, Chinthapenta V , Ramji M*. A novel approach to analyze adhesive layer strain field in a stepped lap repaired carbon fiber reinforced polymer panel using digital image correlation. Journal of Adhesion Science and Technology, 31(19-20), 2180-2201, 2017. doi.org/10.1080/01694243.2016.1264660	3	5.6
8	Yang Lei, Gong Ze, Lin Yuan*, Chinthapenta Viswanath , Li Qunyang, Webster Thomas J, and Sheldon Brian W*. Biomaterials: Disordered Topography Mediates Filopodial Extension and Morphology of Cells on Stiff Materials. Advanced Functional Materials, 27 (38), 2017. doi.org/10.1002/adfm.201702689	19	27.7
7	Murshid Imam*, Vikrant Racherla, Kajal Biswas, Hidetoshi Fujii, Viswanath Chintapenta , Yufeng Sun, and Yoshiaki Morisada. Microstructure-property relation and evolution in friction stir welding of naturally aged 6063 aluminium alloy. The International Journal of Advanced Manufacturing Technology, 91, 1753-1769, 2017. doi.org/10.1007/s00170-016-9865-9	3.5	5.9
6	Kallekar Laxman, Viswanath Chinthapenta , Anand Mohan*. Effect of Wall Flexibility on the Deformation during Flow in a Stenosed Coronary Artery, Fluids, 2(2),16, 2017.doi.org/10.3390/fluids2020016	1.9	4
5	Somashekara MA, Naveenkumar M, Kumar Avinash, Viswanath C , and Simhambhatla S*; Investigations into the effect of weld-deposition pattern on residual stress evolution for metallic additive manufacturing, The International Journal of Advanced Manufacturing Technology, 90, 2009-2025, 2017. doi.org/10.1007/s00170-016-9510-7	3.5	5.9
4	Khedkar Sourabh, Chinthapenta Viswanath , Madhavan Mahendrakumar, Ramji M*. Progressive failure analysis of CFRP laminate with interacting holes under compressive loading, Journal of Composite Materials, 49(26),3263-3283, 2015. doi.org/10.1177/0021998314561810	2.8	5.1
3	Kothari Abhishek K, Xiao Xingcheng, Sheldon Brian W*, Chintapenta Viswanath . The impact of nanocrystalline diamond grain boundary chemistry on the frictional response in sliding contact with 319Al alloy. Journal of Materials Science, 50(7), 2993-3003, 2015. doi.org/10.1007/s10853-015-8864-9	4.3	7.6
2	Srilakshmi R, Ramji M, Chinthapenta Viswanath *. Fatigue crack growth study of CFRP patch-repaired Al 2014-T6 panel having an inclined center crack using FEA and DIC. Engineering Fracture Mechanics, 134,182-201, 2015. doi.org/10.1016/j.engfracmech.2014.12.012	5.3	8.6
1	L Yang, V Chinthapenta , Q Li, D Stout, A Laing, BW Sheldon, TJ Webster*; Understanding osteoblast responses to stiff nano-topographies through experiments and computational simulations, Journal of Biomedical Materials Research Part A 97 (4), 375-382, 2011. doi.org/10.1002/jbm.a.33094	3.9	9

c) Conference Proceedings/Book Chapters/Journals Without Impact Factor

SNo	Journal Publications/Conference Proceedings	CS
26	A Sabale, MS Rizvi, V Chinthapenta , Avinash Eranki. Ultrasound Wave Propagation and Shear Stress through Tissues for Intercostal Focused Ultrasound Therapy. 2024 IEEE South Asian Ultrasonics Symposium (SAUS), 1-4, 2024. doi.org/10.1109/SAUS61785.2024.10563729	-
25	R Mishra, M Imam, V Chinthapenta . Microstructure and Mechanical Properties of Wire Arc Additively Manufactured Inconel 625 Superalloy. Recent Advances in Additive Manufacturing, Volume 2: Select Proceedings of ICAM 2024. doi.org/10.1007/978-981-96-3165-0_28	0.9
24	VS Abhiram, Mathews Peter, VS Anaswara, Premdip Kumar, Sumant Yadav, Dharmendra Kumar, M Jacob Koshy, KP Vineesh, Murshid Imam, Viswanath Chinthapenta , Raghuram, Karthik Desu, J John Rozario Jegaraj. Coupled CFD-DEM Numerical Modeling of SLM Process of Ti-6Al-4V Alloy Using Temperature-Dependent Properties. International Conference on Additive Manufacturing. Pp. 427-438, 2024. doi.org/10.1007/978-981-97-6016-9_33	0.9

23	Rajnish Mishra, Murshid Imam, Viswanath Chinthapenta , P Mastanaiah. Thermo-mechanical modelling of the wire arc-based additively manufactured Inconel 625 superalloy. <i>Materials Today: Proceedings</i> , 108, pp. 53-64, 2024. doi.org/10.1016/j.matpr.2023.08.142	6.6
22	Gopi Gulivindala, Rakesh Kumar Mothukuru, Madhukiran Karanam, Kwong Ming Tse, Viswanath Chinthapenta . Yield locus of FCC single crystal for biaxial loading. <i>Materials Today: Proceedings</i> , 108, pp. 135-140, 2024. doi.org/10.1016/j.matpr.2024.01.050	6.6
21	S Chalasani, S Potukuchi, S Rayasam, V Narayanamurthy, V Chinthapenta . Mechanical characterization of high-strength carbon-epoxy composite laminates. <i>Materials Today: Proceedings</i> , 108, pp 80-85, 2024. doi.org/10.1016/j.matpr.2023.09.092	6.6
20	MA Wahed, M Imam, V Chinthapenta , E Jimenez-Melero, MAA Anshari. Additive friction stir processing and hybrid metal additive manufacturing of high melting point materials: A review. <i>Materials Today: Proceedings</i> , 2023. doi.org/10.1016/j.matpr.2023.08.018	6.6
19	Tippanna, R Mishra, M Imam, V Chinthapenta , KP Vineesh. Temperature evolution in WAAM process without overlapping weld beads. <i>American Institute of Physics Conference Series</i> 2745 (1), 030002, 2023. doi.org/10.1063/5.0143318	0.5
18	H Gururani, SNSH Chittajallu, M Doulatramani, M Ramji, S Basu. Implications of Graft-Host Structural Misalignment during Penetrating Keratoplasty on Topographical Irregularities. <i>Investigative Ophthalmology & Visual Science</i> 64 (8), 5142-5142	-
17	D Kumar, OP Rahate, M Imam, J John Rozario Jegaraj, V Chinthapenta . Modeling of Inconel Powder Flow and Thermal Distribution During the Direct Energy Deposition Process. <i>Advances in Additive Manufacturing and Metal Joining: Lecture Notes in Mechanical Engineering</i> , 2023, pp. 105–112. <i>Proceedings of AIMTDR</i> . 2023. doi.org/10.1007/978-981-19-7612-4_9	0.9
16	A Sabale, MS Rizvi, V Chinthapenta , Avinash Eranki. Effect of Focused Ultrasound on Shear Wave Propagation in Nonlinear Media. <i>Lecture Notes in Mechanical Engineering</i> , pp 219-228, 2024. doi.org/10.1007/978-981-97-6367-2_19	0.9
15	SNSH Chittajallu, H Gururani, A Richhariya, KM Tse, V Chinthapenta. Strength characterization of Human Cornea Against Sutures: Experiments. <i>Investigative Ophthalmology & Visual Science</i> 63 (7), 98–A0196-98–A0196	-
14	H Gururani, SNS Harsha Chittajallu, A Richhariya, M Ramji, Sayan Basu, Viswanath Chinthapenta. Investigation on rabbit corneal birefringence using digital photoelasticity technique. <i>Investigative Ophthalmology & Visual Science</i> 63 (7), 1203–A0203-1203–A0203	-
13	G. Vamsi Krishna, V. Narayanamurthy, and C. Viswanath . Effect of Longitudinal C Unstiffened Cut-outs on Buckling Behavior of FRP Strengthened Metallic Airframe Shell Structures. <i>Lecture Notes in Mechanical Engineering</i> , 2022, pp. 423–430 <i>Proceedings of VSAM 2021</i> , 29-30th, May 2021. doi.org/10.1007/978-981-16-9539-1_30	0.9
12	G Gulivindala, MK Karanam, V Chintapenta . Development of Digital Twin for Polycrystalline BCC α -Fe, Recent Advances in Applied Mechanics, 2022, 535-545. doi.org/10.1007/978-981-16-9539-1_39	0.9
11	Amit Chaudhary, Rajnish Mishra, Anwar Ali Anshari, Murshid Imam, Chinthapenta R Viswanath . Experimental and Numerical Investigation of Laser-FSW Hybrid Welding Technique for High Strength Materials. <i>Materials Today: Proceedings</i> , 65, pp. 3437-3448, 2022. doi.org/10.1016/j.matpr.2022.05.579	6.6
10	Murshid Imam, Sai Naga Sri Harsha Chittajallu, Himanshu Gururani, Hajime Yamamoto, Kazuhiro Ito, Pradeep Kumar Parchuri, Rajnish Mishra, Ashutosh Richhariya, Viswanath Chinthapenta . Experimental study on improving the additively manufactured GMAW and TIG beads using FSP. <i>Materials Today: Proceedings</i> , 2022, 56, pp. 690–705. doi.org/10.1016/j.matpr.2022.01.154	6.6
9	G. Vamsi Krishna, C. Viswanath , and Ashok Kumar Pandey. Performance of Bolted Joint Modelling Using Master Element. <i>Lecture Notes in Mechanical Engineering</i> , Pages 595 – 605. 2021, 6th National Symposium on Rotor Dynamics, NSRD 2019. doi.org/10.1007/978-981-15-5701-9_48	0.9
8	Amit Chaudhary, Chiranthan Ramesh, Viswanath Chinthapenta , and Murshid Imam. Tool Durability and Weldability in Hybrid Friction Stir Welding of High-Strength Materials. <i>Lecture Notes on Multidisciplinary Industrial Engineering</i> , 2019, Part F257, pp. 145–152. doi.org/10.1007/978-981-32-9072-3_13	1.1

7	T Chaitanya Sagar, V Chinthapenta . Lower and upper bound estimates of material properties of pristine graphene: Using Quantum Espresso. Advances in Structural Integrity: Proceedings of SICE 2016, 253-265. doi.org/10.1007/978-981-10-7197-3_22	-
6	Sourabh Khedar, Viswanath Chinthapenta , CM Manjunatha, M Ramji. Progressive damage study in an adhesively bonded patch repaired open hole CFRP panel under compressive loading. Journal of Aerospace Sciences and Technologies 67 (2B), 299-308. joast.org/index.php/joast/article/view/351	-
5	L Yang, Q Li, V Chinthapenta , A Laing, BW Sheldon, TJ Webster; The impact of material nanotopography on cell functions and filopodia extension: Experiments and modeling, <i>Materials Research Society Symposium Proceedings</i> , 2010, 1236, pp. 138–143. doi.org/10.1557/PROC-1236-SS10-08	-
4	C Viswanath R. R. S. R. , DV Kubair. Boundary layer analysis of mode-3 crack in ductile radial functionally graded materials. ICCES05-Madras	-
3	B Bhanuchandar, C Viswanath R. R. S. R. , DV Kubair. Spontaneous crack propagation in bi-material functionally graded materials. ICCES05-Madras	-
2	SH Krishna, V Chinthapenta , HN Mohan Raghavan. Finite Element Modelling Of an Anatomically Accurate Human Spinal Cord. doi.org/10.11159/icbes24.110	-
1	GV Krishna, V Narayanamurthy, C Viswanath . Buckling behavior of FRP strengthened cylindrical metallic shells with initial geometric imperfections. Journal of Structural Engineering (Madras)	-

d) Publication under Review:

SNo	Details
4	S. Dhiman, M. Brandt, D. Fabijanic, V. Chinthapenta*, W. Xu*, Controlling microstructure with tailorable mesoscale spatial heterogeneity in additively manufactured Ti-6Al-4V via laser powder bed fusion, (submitted to Acta Materialia).
3	S. Dhiman, Yixuan Wang, M. Brandt, D. Fabijanic, V. Chinthapenta*, W. Xu*, Evolution of bimodal microstructure in laser powder bed fusion fabricated Ti-6Al-4V during cyclic heat treatments, Scripta Materialia (Submitted to Scripta Materialia)
2	Kondababu Kadali, Durgaprasad Kadambari, Venkatesh Narise, Manish Tak, Bernard Rolfe, Wei Xu*, Viswanath Chinthapenta. Enhanced Melt-Track Characteristics of Refractory Tungsten From Preheat Assisted Electron-beam Surface Melting: An Experimental and Numerical Investigation. (Submitted to Acta Materialia)
1	Rasmiranjan D, Vinay S, Sai NSHC, Vineet J, Ramji M, Sayan B, Viswanath C. Experimental Investigation of Tensile Properties and Fracture Behavior of Ophthalmic Sutures under Straight, Knotted, and Looped Configurations: Vicryl, Nylon, and Polypropylene (Submitted to PLOS One)

e) Publications near to submission stage:

SNo	Details
3	Sai Naga Sri Harsha Chittajallu, Himanshu Gururani, Rasmiranjan Dalai, Saggam Vinay Kumar, Subha Narayan Rath, Sayan Basu, Viswanath Chinthapenta. Suture Retention Nature of The Human Cornea
2	Himanshu et al. Post-penetrating keratoplasty astigmatism from a biomechanical viewpoint
1	Sasidhar et al. Fracture analysis in Hydrogels under Flexural loading

Projects:

a) Project Funding as PI: 772.375 Lakhs

SNo	Status	PI	Co-PI	Period	Agency	Amount in Lakhs	Title
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13	Ongoing	VC	Dr Sai S	2024-25	CARS	19.96	Thermo-Structural Analysis of C-C brakes
12	Ongoing	VC	Dr Sayan Basu, Prof Ramji	2024-27	DBT	209.64	A device based on digital photoelasticity for in vivo characterization of corneal birefringence
11	Ongoing	VC	-	2023-25	CARS	9.94	Study of Constitutive modeling of Graphite for Thermal and pressure Loads
10	Ongoing	VC	Dr Syed, Dr Rajesh	2023-26	GTRE	213.6	Numerical modeling of SLM process for 3D printing of Inconel and Ti-6-4 alloys
9	Ongoing	VC	Dr Sayan Basu	2023-26	CRG	29	Corneal Characterization under biaxial-loading in physiological conditions
8	Completed	VC	Prof Ramji, Dr Rajesh Korla	2024-25	BorgWarner	21.33	Accelerated Creep Modeling and Experiments on 1.4848 Carbon Steel
7	Completed	VC	Dr Imam	2023-25	DRDL	87.465	Direct Metal Laser Sintering of C103 Refractory Alloy
6	Completed	VC	-	2017-18	GIAN	5.44	Integrated Computational Materials Engineering Foreign Faculty Prof Horstemeyer
5	Completed	VC	-	2019-20	La Foundation	12	Biomechanical Simulations of Resuscitation
4	Completed	Dr Murshid and VC	Prof Roger, Prof Enrique	2019-22	SPARC + UKIERI	54	Hybrid 3D printing with GMAW-twin wire-based additive layer enhanced by friction stir processing
3	Completed	VC	Dr Ashutosh	2018-23	LVPEI	79	Characterization of Bio-Synthetic Cornea
2	Completed	VC	-	2017-20	ECR	28	Void Growth Modelling in BCC Steels
1	Completed	VC	-	2016-17	Seed IITH	3	Homogenization of non-local RVE

b) Project Funding as CoPI: 771.44 Lakhs

SNo	Status	PI	Co-PI	Period	Agency	Amount in Lakhs	Title
12	Ongoing	Prof Ramji	VC	2025-27	DBT	35.51	Novel neck screw design with miniplate for fixation of mandibular condylar neck fractures- An evaluation using 3-Dimensional finite element analysis/ 3D-Digital Image Correlation
11	Ongoing	Dr Rajesh	VC	2024-27	DMRL	552.5	Optimization of Electron Beam AM Process of Ti-6Al-4V to minimize the anisotropy in high-temperature mechanical properties, creep, fatigue, and

							fatigue crack growth demonstrate printing of real-time component with optimized process parameters.
10	Ongoing	Dr Prabhat	Dr Sai, VC	2025	Hyundai	15	Bending stiffness for the truck chassis with variable stiffeners
9	Completed	Prof Ramji	Prof Gangadharna and VC	2024-25	CARS	50	Acceptance criterion of the Composite Pressure Vessels involving progressive damage modelling
8	Completed	Prof Gangadharan	Dr Sai, VC	2024	Hyundai	17	Windshield Vibrational analysis
7	Completed	Dr Prabhat	Dr Sai, VC	2024	Hyundai	19	Optimization of the size and shape of the vanes using ML methods
6	Completed	Prof Gangadharan	Dr Sai, VC	2023-24	SAARC	1.5	Random vibration analysis of SAARC cases
5	Completed	Dr Raghu	Mentor VC	2019-20	NPDF	19	National Post-Doctoral Fellowship
4	Completed	Prof Sushmee	VC	2019-20	ATAL FDP	0.93	Nanomaterials and Nanomechanics and its applications in devices and sensors
3	Completed	Prof Sushmee	VC	2018-24	TL-IITH	47	Tinkerer Lab establishment sponsored by IITB Alumnus
2	Completed	Himanshu	Mentor VC	2018-22	PMO-Fellowship	0	PMO-Fellowship
1	Completed	Prof Ashok	VC	2014-15	AR&DB	14	Static and Dynamic Characterization of Bolted Structures

c) Proposals Submitted:

SNo	Status	PI	Co-PI	Period	Agency	Amount (L)	Title
3	Under Revision	VC	Dr Gopi and Prof Prem HCU	2025-27	GTMAP	400	LSP of Curved Components
2	Under Revision	Dr Sai	VC	2025-27	BRNS	30	Modeling and Simulation of Ballooning in Fuel Clad Tubes
1	Under Revision	Dr Sai	VC and Prof Ramji	2025-27	CARS	50	Wear Analysis of C-C brakes

Peer-Review Contributions:

a) Journals

ACS Omega	Heliyon	Micromechanics and Molecular Physics
Additive Manufacturing	High-Temperature Materials and Processes	Nondestructive Testing and Evaluation

Additive Manufacturing Frontiers	IEEE Sensors	Optics and Laser Technology
Advanced Healthcare Materials	INAE Letters	PLOS ONE
Advances in Engineering Sciences	International Journal of Fracture	Polymer Composites
Advances in Engineering Sciences and Applied Mathematics	Journal of Intelligent Manufacturing	Polymers
Alloys and Compounds	Langmuir	Recent Progress in Materials
Applied Sciences	Materials	Sadhana
Arabian Journal for Science and Engineering	Materials & Design	Smart Materials and Structures
BioEngineering	Materials Engineering and Performance	Steel and Composite Structures
Biomechanics and Modeling in Mechanobiology	Materials Performance and Characterization	Structural Control and Health Monitoring
Composite Interfaces	Materials Research	Structural Health Monitoring
Computational Methods in Engineering Science & Mechanics	Mathematics	Surface & Coatings Technology
Computers and Structures	Mechanical Engineering Science	Theoretical and Applied Fracture Mechanics
Defence Science Journal	Mechanics of Advanced Materials and Structures	Thermal Stresses
Electronics	Mechanics of Materials	Thin-Walled Structures
Engineering Failure Analysis	Mechanics-Based Design of Structures and Machines	Transactions of IIM
Engineering Fracture Mechanics	Metallurgical and Materials Transactions A	Transactions of INAE
Engineers Australia Technical Journals	Metals and Materials International	Transactions on Biomedical Engineering

b) Conference Reviews:

- iNaCoMM 2025
- ICEAMS-2025
- ISSC 2020
- GYTI-2018
- IHMTC-2017
- COPEN-2017
- IJMSI-2017
- PLMSS-2015

c) Thesis/Proposals/Other Reviews:

- IITM-MTech Thesis review: Analysis of large deformations in stimuli-responsive polymeric thin films
- IISc-PhD Thesis review: Role of Crosslinks and Fiber Reinforcement in Hydrogel Mechanics
- IITBHU-PhD Thesis: Numerical modeling of fatigue behaviour of Polycrystalline materials
- Fondecyt-Chile: Biomechanical analysis of soft tissue behaviour
- IITD-PhD Thesis: Coupled Magneto-Elastic and Electro-Elastic effects in rods
- IITDh-PhD Thesis: Design and development of 4D printing technique with multi-materials and multifunctionality
- GIAN-IIT-KGP
- PMRF-Reviews

Phd Students:

a) Ongoing PhD Students

SNo	Student Name	Thesis title	Program	Guide	Co-Guide	Status
23	Rahul K S	NDT of AM	IITH-Regular	VC	Dr Tulsiram	1st Year, Joined Aug 2025
22	Tatyaso A G	Micromechanics of Graphite	IITH-Regular	VC	Dr Sai S	1st Year, Joined Aug 2025
21	Mahesh P	Ultrasound NDT	IITH-Regular	VC	Dr Tulsiram	2nd Year, Joined Aug 2024
20	Manindra	Optometric investigation of Corneal Thinning	IITH-LVPEI	VC, Dr Sayan Basu	-	1st Year, Joined Dec 2024
19	U Kapil Choudhary	Digital Photoelasticity	IITH-LVPEI	VC, Dr Sayan Basu	-	2nd Year, Joined Aug 2024
18	N Venkatesh	Full-scale AM component development using Simulations	IITH-Regular	VC	-	2nd Year, Joined Aug 2022
17	K V Durga Prasad	Creep simulations of AM Tungsten	IITH-Deakin	VC, Prof Wei Xu	Dr Rajesh K	3rd Year, Joined Aug 2023
16	Vinay Kumar S	Computational Damage Modelling	IITH-Regular	VC	Prof Gangadharan	3rd year, Joined Aug 2023
15	Rakesh Kumar M	Shock properties of Tungsten/Tantalum	IITH-External	VC	Drs Madhu, Phani	3rd year, Joined Dec 2023
14	Rasmiranjan Dalai	Biaxial Testing of Cornea	IITH-LVPEI	VC, Dr Sayan Basu	Prof Ramji	3rd Year, Joined Aug 2023
13	Kondababu Kadali	AM feasibility of Tungsten	IITH-Deakin	VC, Prof Wei Xu	-	4th Year, Joined Aug 2022
12	Abdul Khalid	AI-ML optimization of 3D printed Inconel alloys	IITH-Deakin	VC, Prof Wei Xu	-	4th Year, Joined Aug 2021
11	M Muneer	Numerical simulations of coagulation in the flexible arteries	IITH-External	VC	Prof Anand	4th Year, Joined Aug 2021
10	Mahesh P	Fatigue and fracture of composite structures	IITH-Regular	Prof Ramji	VC	5th Year, Joined Aug 2020
9	Aryendra Singh	Modeling of Laser Shock Peening	IITH-DMRL	VC	Dr Phani	Thesis submission for OC by Nov 2025
8	Ch Suryakiran	Composite shim Design, fabrication, and analysis.	IITH-External	VC	Dr Vijayabhaskar	Thesis submission for OC by Nov 2025

b) Graduated Phd Students

SNo	Student Name	Thesis title	Program	Guide	Co-Guide	Status
7	Sahil Dhiman	Engineering 3D printed titanium alloys	IITH-Deakin	VC, Prof Wei Xu	-	OC completed on 17th Oct 2025
6	Sri Naga Harsha	Mechanical stability analysis of donor and Bio-synthetic Cornea	IITH-LVPEI-SUT	VC, Dr KM	Dr Sayan Basu	Graduated on Jan 10 2025; Working in DreamVu Startup

5	G Gopi	Void growth and coalescence studies in BCC metals	IITH-SUT	VC, Dr KM	-	Graduated on Apr 4, 2024, Now PDoc at IITB
4	Himanshu G	Optical stability analysis of the human Cornea	IITH-LVPEI	VC	Dr Sayan Basu, Prof Ramji	Graduated on Feb 1, 2024, Working at DSS Pune
3	G Vamsi Krishna	Strengthening of the shell using composite materials	IITH-External	VC	Dr Vijayabhaskar	Graduated in Sep 2021, Working in DRDO
2	Madhu Kiran K	Void growth studies in FCC metals	IITH-External	VC	-	Graduated in May 2021, Working in DRDO (Lateral entry in 2025)
1	T Chaitanya Sagar	Multi-Scale Mechanics of Graphene and Stanine	IITH-Regular	VC	-	Graduated in Feb 2021, Working in a Startup, USA

Post-Doctoral Fellow:

SNo	Student Name	Project	Year	Remarks
4	Dr Sasidhar P	Fracture behavior of micro/nano-particle-based hydrogel	2021-23	Now in Alstrom
3	Dr. T Chaitanya S	Fracture toughness of 2D materials	2021-22	USA Startup
2	Dr. Alok Behra	High-Temperature Composites Characterization	2020-21	Now in IISc
1	Dr. Raghu Ram	Wave propagation in granular crystals using discrete element modeling	2020-20	Now Asst Prof NIT Tiruchunapally

MTech Students:

SNo	Student Name	Thesis Topic	Year	Role
37	Om B	Fracture Modelling of Sutures	2026	Guide
36	Shivam	Modelling of Sutures	2026	Guide
35	Rajnish Kumar Yadav	Modelling of CC Brake	2025	Guide
34	Vinayak Maharshi	Triaxial loading of Graphite	2025	Guide
33	Reddi babu	Laser Particle Interaction in DED	2025	Guide
32	Moggala Sri Charan	Hard Tool Coatings using WAAM	2024	Guide
31	Boddepalli Pavan	Random Vibration Simulations	2024	Guide
30	Akshat	Thermo-Mechanical Analysis of Multi-Stacked IC	2024	Guide
29	Nishanth	Studies on an efficient cooling system of 2.5D	2024	Guide
28	Logesh Krishna K	Design and Analysis of Heat Exchanger Structure	2023	Guide
27	Sarath Chandra	Mechanical behaviour of a new high temperature nickel-based superalloy, Inconel 939	2023	Guide
26	Gunda Sachin	Modelling, simulations, and experiments of fibre-reinforced dielectric polymers.	2022	Guide
25	Satya Pradhan	Electro-elastic Deformations in Thin Shells	2022	Guide
24	K Abhitej	Topological optimization of AM Pipe connections	2021	Guide
23	Popuri Shiri Lekhaz	Modeling of CMCs	2021	Guide
22	Potti Kartheek Durga	Micromechanics of composites	2021	Guide
21	Shepal B	Special Singular elements	2021	Co-Guide
20	Deep Saparia	Micromechanics of composites	2020	Guide
19	Sridath	Hybrid Additive Manufacturing: GMAW+ FSP	2019	Guide

18	P Venkatadri	Crystal plasticity studies of BCC metals with inhomogeneity using size effects	2018	Guide
17	Rajeev Jain	Study of interface delamination in unidirectional CFRP laminate under cyclic loading conditions: Experiments and Cohesive zone modelling	2018	Guide
16	T Kishor	Fracture Studies in Biomaterials	2018	Guide
15	Saurabh Soni	Fracture Studies of Surface Cracks	2018	Guide
14	S Rajeswar	Smoldering Experiments and Simulations of an Incense Stick	2017	Guide
13	Rahul K	Numerical simulations of laser-assisted friction stir welding	2017	Guide
12	Srimaya padhi	Dynamic Failure Analysis in Isotropic and Anisotropic Bi-material Interface	2017	Guide
11	Pankaj Pandya	Investigation of Single Fiber Composite residual strength pre and post fragmentation: Using Progressive Damage Analysis and HFGMC framework	2017	Guide
10	Pradeep P	A Study of Melt Ratio and Microstructure in Dissimilar Fusion Welds	2016	Co-Guide
9	Vinayak Mathur	Heat Transfer Modelling of AC Waveforms Submerged ARC Welding Process	2016	Co-Guide
8	Imran Khan	Modeling of the temperature profile for the plunge stage of the friction stir welding process	2016	Co-Guide
7	Vamsi Krishna	Dynamic Behavior of Structures with Bolted Joints	2016	Co-Guide
6	Brijesh Patel	Progressive Damage Analysis of Open Hole Carbon Fiber Reinforced Polymer Panel under Tensile Loading	2016	Co-Guide
5	Mohini Kale	Investigation of Minimum RVE Size for CFRP Laminates: HFGMC and PDA Framework	2016	Guide
4	Sukanya Joshi	Study of interface delamination in unidirectional CFRP laminate under static loading conditions: Experiments and Cohesive zone modeling	2016	Guide
3	H Sriram	Cohesive zone modelling of cracked interface	2015	Guide
2	Rajesh Kumar Meena	Micro-mechanics of Re-entrant Hexagonal Auxetic Structures and Their Blast Resistance	2015	Guide
1	Jistom	Evaluation of fracture parameters from XFEM and experimental displacement fields	2014	Guide

Courses taught (Average 12.14 Credits per year):

SNo	Acad Year	Academic Period	Course Code	Course Name	Credits	Instructor(s)	No of Students Registered	No of Students submitted feedback	% Response	Score
54	2024-25	JAN-MAY 2025	ME5690	Advanced FEM	3	Dr. Viswanath Chinthapenta	31	23	74.19	4.19
53	2024-25	JUL-NOV 2024	ME5130	Finite Element Method	3	Dr. Viswanath Chinthapenta	67	64	95.5	4.41
52	2023-24	JUL23-NOV23	ME5769	Applied Solid Mechanics	1.5	Dr. Viswanath Chinthapenta	5	2	40	4.20
51	2023-24	JUL23-NOV23	IP5010	Fatigue and Fracture in Electronic Packaging	3	Dr. Viswanath Chinthapenta	5	4	80	4.60
50	2023-24	JAN-MAY 2024	ME5690	Advanced FEM	3	Dr. Viswanath Chinthapenta	39	35	89.7	3.80
49	2023-24	JAN-MAY 2024	OP5110	Computational & Numerical Methods	2	Dr. Viswanath Chinthapenta	7	5	71.4	4.60
48	2023-24	JAN-MAY 2024	IC5143	Computational Micromechanics	3	Dr. Viswanath Chinthapenta	7	5	71.4	4.64
47	2022-23	OCT22-FEB23_UG I SEM	OP5110	Computational & Numerical Methods	2	Dr. Viswanath Chinthapenta	3	3	100	4.07

46	2022-23	OCT22-FEB23 _UG I SEM	ME5799	Topics in Computational Mechanics	3	Dr. Viswanath Chinthapenta	3	3	100	4.27
45	2022-23	JAN23-MAY23	OP5110	Computational & Numerical Methods	2	Dr. Viswanath Chinthapenta	3	3	100	4.07
44	2022-23	JAN23-MAY23	ME5799	Topics in Computational Mechanics	3	Dr. Viswanath Chinthapenta	3	3	100	4.27
43	2022-23	AUG22-NOV22	OP5133	Stress Analysis & Biomechanics	2	Dr. Viswanath Chinthapenta	4	3	75	3.33
42	2022-23	AUG22-NOV22	IS5030	Computational Modelling Techniques	2	Dr. Viswanath Chinthapenta	16	10	62.5	4.00
41	2022-23	AUG22-NOV22	IP5010	Fatigue and Fracture in Electronic Packaging	3	Dr. Viswanath Chinthapenta	7	7	100	4.29
40	2022-23	AUG22-NOV22	MS5523	Introduction to ICME Techniques and Tools	3	Dr. Viswanath Chinthapenta	5	4	80	5.00
39	2021-22	MAR22-JUL22 UG IISEM	ME1020	Engineering Mechanics	3	Dr. Viswanath Chinthapenta	226	200	88.45	4.08
38	2021-22	JAN22-APR22	ME1020	Engineering Mechanics	3	Dr. Viswanath Chinthapenta	226	200	88.45	4.08
37	2021-22	AUG21 - NOV21	ME5130	Finite Element Method	3	Dr. Viswanath Chinthapenta	84	74	88.04	4.17
36	2021-22	AUG21 - NOV21	IS5030	Computational Modelling Techniques	2	Dr. Viswanath Chinthapenta	21	21	98.81	4.26
35	2020-21	NOV20-FEB21 _UG I SEM	ME1020	Engineering Mechanics	3	Dr. Viswanath Chinthapenta	109	108	98.26	3.78
34	2020-21	NOV20-FEB21 _UG I SEM	ME1030	Dynamics	2	Dr. Viswanath Chinthapenta	24	23	95.83	3.84
33	2020-21	NOV20-FEB21 _UG I SEM	ID1130	Engineering Statics	2	Dr. Viswanath Chinthapenta	13	13	100	4.10
32	2020-21	AUG20 - NOV20	ID2020	Solid Mechanics - II	2	Dr. Viswanath Chinthapenta	84	84	98.45	3.15
31	2020-21	AUG20 - NOV20	ME3150	Applied Elasticity	2	Dr. Viswanath Chinthapenta	54	54	97.04	3.18
30	2020-21	AUG20 - NOV20	ID1160	Solid Mechanics - I	2	Dr. Viswanath Chinthapenta	7	7	100	3.39
29	2020-21	AUG20 - NOV20	ME1020	Engineering Mechanics	3	Dr. Viswanath Chinthapenta	109	108	98.26	3.78
28	2020-21	AUG20 - NOV20	ME1030	Dynamics	2	Dr. Viswanath Chinthapenta	24	23	95.83	3.84
27	2020-21	AUG20 - NOV20	IS5030	Computational Modelling Techniques	2	Dr. Viswanath Chinthapenta	12	11	91.67	3.94
26	2020-21	AUG20 - NOV20	ID1130	Engineering Statics	2	Dr. Viswanath Chinthapenta	13	13	100	4.10
25	2020-21	AUG20 - NOV20	CC5533	Advanced Computing in Climate Change	1	Dr. Viswanath Chinthapenta	10	10	96	4.23
24	2019-20	JAN20 - APR20	ME5610	Fracture Mechanics	3	Dr. Viswanath Chinthapenta	11	11	99.09	4.11
23	2019-20	AUG19 - NOV19	ME3080	Design of Machine Elements	2	Dr. Viswanath Chinthapenta	49	48	97.24	3.93
22	2019-20	AUG19 - NOV19	ME5130	Finite Element Method	3	Dr. Viswanath Chinthapenta	52	50	95.87	3.97
21	2019-20	AUG19 - NOV19	CC5533	Advanced Computing in Climate Change	1	Dr. Viswanath Chinthapenta	15	14	93.33	4.21
20	2018-19	MAY18 - JUL18 - Summer	ID2020	Solid Mechanics - II	2	Dr. Viswanath Chinthapenta	12	1	8.33	4.20
19	2018-19	MAY18 - JUL18 - Summer	ME5750	Micro-Mechanics of Defects	3	Dr. Viswanath Chinthapenta	5	2	40	4.50

18	2018-19	JAN19 - APR19	ME2080	Introduction to Mathematical Modelling	1	Dr. Viswanath Chinthapenta	80	11	13.25	4.32
17	2018-19	JAN18 - APR18	ME5610	Fracture Mechanics	3	Dr. Viswanath Chinthapenta	17	9	52.94	4.79
16	2018-19	AUG18 - NOV18	ME3445	Finite Element Methods Lab	1	Dr. Viswanath Chinthapenta	48	2	3.85	2.46
15	2018-19	AUG18 - NOV18	ME5130	Finite Element Method	3	Dr. Viswanath Chinthapenta	81	32	39.32	4.46
14	2017-18	AUG17 - NOV17	ME5260	Continuum Mechanics	3	Dr. Viswanath Chinthapenta	39	17	43.59	3.59
13	2017-18	AUG17 - NOV17	ID2020	Solid Mechanics - II	2	Dr. Viswanath Chinthapenta	76	20	26.32	3.87
12	2016-17	JUL16 - NOV16	ME3080	Design of Machine Elements	2	Dr. Viswanath Chinthapenta	42	2	4.76	3.65
11	2016-17	JUL16 - NOV16	ME3150	Applied Elasticity	2	Dr. Viswanath Chinthapenta	48	2	4.17	4.23
10	2016-17	JUL16 - NOV16	ME3090	Design of Transmission Elements	2	Dr. Viswanath Chinthapenta	42	1	2.38	4.85
9	2016-17	JAN17-APR17	ME3413	Machine Drawing & Solid Modelling	2	Dr. Viswanath Chinthapenta	42	7	16.67	3.61
8	2016-17	JAN17-APR17	ME5610	Fracture Mechanics	3	Dr. Viswanath Chinthapenta	28	11	39.29	4.18
7	2015-16	JUL15 - NOV15	ME3130	Design of Machine Elements	4	Dr. Viswanath Chinthapenta	51	16	31.37	3.50
6	2015-16	JAN16 - APR16	ME3413	Machine Drawing & Solid Modelling	3	Dr. Viswanath Chinthapenta	50	6	12	3.53
5	2015-16	JAN16 - APR16	ME5610	Fracture Mechanics	3	Dr. Viswanath Chinthapenta	23	12	52.17	4.53
4	2015-16	JAN16 - APR16	ME5690	Advanced FEM	3	Dr. Viswanath Chinthapenta	33	9	15.15	4.62
3	2014-15	JAN15 - APR15	ME5690	Advanced FEM	3	Dr. Viswanath Chinthapenta	30	19	63.17	3.89
2	2014-15	AUG14 - NOV14	ME2110	Mechanics of Solids	4	Dr. Viswanath Chinthapenta	91	61	66.32	3.97
1	2014-15	AUG14 - NOV14	ME5130	Finite Element Method	3	Dr. Viswanath Chinthapenta	63	55	86.98	4.26

Workshops Conducted/Resource Person:

S No	Year	Role	Event
21	2025	Co-Convenor, IITH	ICEAMS, 2025
20	2025	Invited Talk	PM-25
19	2023	Invited Talk	ICONS-2023
18	2022	Convenor	International Conference SICE-2022
17	2021	Speaker, Coordinators: Dr. Sushmee and Dr. Viswanath	ATAL FDP on Nanomaterials and Nanomechanics and its applications in devices and sensors
16	2021	Speaker	ATAL FDP on Simulation of Welds and Optimization Techniques
15	2021	Speaker	NPIU workshop on "3D Printing and Design", Coordinator Dr. Onkar
14	2020	Speaker	TEQIP: Aid of Demo Experiments in Teaching Solid Mechanics by Prof Ramji
13	2020	Coordinators: Dr. Imam Murshid, IITPatna, Dr. Viswanath IITH, Dr. Enrique, U Manchester	SPARC Indo-UK: Metal Additive Manufacturing And Friction Stir Processing: Present and Future

12	2019	Speaker, Coordinators: Dr. Viswanath, Dr. Eswar Prasad (IITI), Prof. Mark Horstemeyer	GIAN: Integrated Computational Materials Engineering for Metals
11	2019	Coordinators: Dr. Viswanath and Dr. Gangadharan	TEQIP: Theory and Concepts of Finite Element Methods
10	2019	Coordinator and Speaker	ANSYS workshop from Entuple Technologies
9	2018	Coordinator and Speaker	TEQIP: Theory and Concepts of Finite Element Methods
8	2017	Speaker, Coordinators: Dr. Viswanath and Dr. Syed N K	TEQIP: Theory and Concepts of Finite Element Methods
7	2017	Speaker	TEQIP-II Sponsored Structural Steel Design Workshop; Talk: Nonlinear FEM on steel structures
6	2016	Speaker	TEQIP: Concepts and applications of Finite Element Methods
5	2015	Speaker	ISPAT; Talk: Nonlinear FEM on steel structures
4	2017	Speaker	TLC: Peer Learning
3	2019	Speaker	TLC: Content Development and digital medium
2	2018	Speaker	TLC: UGC: University Act, Statutes and Ordinances
1	2019	Speaker	TLC: Communication in Teaching-Learning Process

Administrative Activities:

a) Institute Activities:

SNo	Duration	Duties
12	2024-till date	Faculty & Staff Grievance Committee Member
11	2023-2024	IITH JEE Adv 2024 Chair
10	2022-2023	IITH JEE Adv 2023 Vice Chair (Zonal)
9	2021-2024	Coordinator for IITH SBI Group Health Insurance
8	2021	Volunteer in the Institute's vaccination drives
7	2021	IITH Gymkhana election body member
6	2020-2022	Associate chair HCU
5	2019- 2020	Warden
4	2019-2020	Elan and Nvision Faculty Co-Advisor
3	2018-2024	Tinkerers Lab Co-Incharge
2	2016-2020	Green office member
1	2016-2019	Institute plantation member

b) Department Activities:

SNo	Duration	Duties
6	2024- till date	FSC
5	2014-2015	Admission Committee
4	2019-2021	DUGC
3	2019-2020	Seminar Coordinator
2	2015-2021	CAE Lab In-Charge
1	2013-2016	Department Placement Coordinator

c) Other Roles:

SNo	Duration	Duties
8	Guest Editor	Material Today Proceedings: SICE-2022
7	Guest Editor	IJ Fracture Special Edition on Structural Integrity
6	Associate Guest Editor	Advances in the Structural Integrity: SICE-2020
5	2023- Till date	General Secretary of InSIS
4	2023- Till date	EC member of ISNT Hyderabad Chapter
3	2024- Till date	GITAM Hyderabad: BOS AE
2	2023- Till date	Gudlavalleru Engg College: BOS ME
1	2016-Till date	Geethanjali College of Engineering and Technology: BOS ME

Infrastructure Established:

a) List

S No	Infrastructure	Funding Source
14	ABAQUS, INTEL, LabVIEW	Project
13	HPCC in Lab	Project and Department
12	High-End Computing Workstations	Project and Department
11	Metrology setup	Project and Department
10	Clean Bio room	Project and Department
9	Reflection mode Digital Photoelasticity (preliminary)	Project
8	Transmission mode Digital Photoelasticity	LVPEI, Project
7	Microscope for capturing the deformation	Project
6	Planar Biaxial testing frame	JICA
5	Uniaxial loading Cell 5N, 50N and Piggy Back	Project
4	MMLab setup and remodelling	Project and Department
3	CAELab Workstations and Desktops	Department
2	CAELab Softwares: ANSYS, ABAQUS, INTEL, TECPLOT	Department
1	CAE Lab B405 remodelling	Department

b) Picture of Facilities

